

Nutrient Management

Nutrient and their Requirements

- Effects of Nutrients
- Manuring in Tamil Nadu – TNAU
- Manuring in Karnataka
- Manuring in Kerala
- Fertilizer Requirement of Adult Palms– KERALA (Under Rainfed Condition)
- Time, Frequency and Method of Fertilizer Application
- Biofertilizers
- Vermicomposting
- Coirpith Composting
- Recycling of Palm Waste
- TNAU Coconut Tonic
- Bio-Fertilizer Recommendation
- Organic Recycling

Physiological Disorder

1. Major Deficiency

- Nitrogen (N) Deficiency
- Phosphorus Deficiency
- Boron (B) Deficiency
- Manganese (Mn) Deficiency
- Magnesium (Mg) Deficiency

2. Minor Deficiency

- Sulphur Deficiency
- Iron Deficiency
- Zinc Deficiency
- Calcium Deficiency
- Copper Deficiency

Protection Technologies of Coconut

Nutrient and their Requirements

Essential elements in coconut nutrition

- Of the primary nutrients, potash (K) has been found to be the most important in coconut cultivation, followed by nitrogen (N). There is a general response to the application of K and N; while response to phosphorous (P) is seen only in certain restricted and localized condition.

- Among the secondary nutrients, magnesium (Mg) and chlorine (Cl) have beneficial effects, followed by calcium (Ca), Sulphur(S) and sodium (Na). Among micro-nutrients, zinc (Zn), boron (B) and manganese (Mn) are required under certain restricted conditions.

Effects of Nutrients:

Nitrogen: The supply of nitrogen which is a constituent of plant cells as well as chlorophyll, the green colouring matter of leaves, is of great importance for the rapid development and growth of the trees. Nitrogen promotes the developments of the vegetative parts of the plant especially the leaves and shoots as well as to increase the number of leaves.

Phosphorus: Phosphorus is found especially in leaves and seeds and also in parts of the plants where vigorous cell division is taking place. It plays an important role in root growth and increased yields.

Potassium: Potash has its important and specific functions apart from those, which it jointly performs in association with phosphorus. It regulates the water economy of the plant and so is indispensable for the rational utilization of limited water supplies for the production of the highest possible yields. It enables the plant to withstand drought. Potash is known to help root development, enabling the palm to take up more nutrients from the soil. Since potash is particularly necessary for the formation of sugar, fat, and fibrous material, the coconut palm may be expected to have a high requirement of potash.

Manganese: It aids in the formation of chlorophyll in leaves. It is present mostly in lower leaves and provides favourable growth conditions during younger stages of palm.

Magnesium: Magnesium (Mg) has beneficial effects on the general growth and productivity of palm. Magnesium improved production of more female flowers, high setting percentage and more number of nuts per bunch. It also plays an important role in photosynthesis and greenness of leaves.

Sulphur: Sulphur increases the oil content and reduces N and sugar contents.

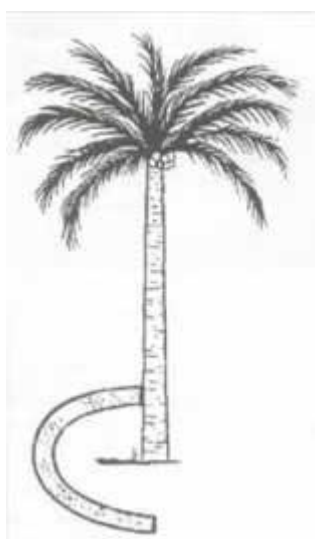
[↑Top of page](#)

Manuring in Tamil Nadu – TNAU

Fertilizer may be applied in two equal splits during June – July and December – January. Apply manures and fertilizers in circular basins of 1.8 m from the base of the palm, incorporate and irrigate sufficient moisture should be present at the time of manuring. Phosphorous may be applied as super phosphate in the basins and incorporated or as DAP through drip when good quality of water is available.

Fertilizer Requirement of Palms – TAMIL NADU

Time after Planting	Compost (kg/palm/year)	Blanket Recommendation (g/palm)			Straight Fertilizer (g/palm)		
		N	P	K	Urea	Super Phosphate	Muriate of Potash
6th month after planting	10	-	-	-	-	-	-
2nd year	20	140	80	300	300	500	500
3rd year	30	280	160	600	610	1000	1000
4th year	40	420	240	900	910	1500	1500
5th year	50	560	320	1200	1300	2000	2000



Manuring in semicircular basins – old method Source: Adhisaya maram thennai(2002)

Time after Planting	DAP & Others (g/palm)			17:17:17: Complex (g/palm)			20:20:0 complex (g/palm)		
	DAP	Urea	MOP	17:17:17 Complex	Urea	MOP	20:20:0 complex	Urea	MOP

2nd year	174	235	500	470	130	365	400	130	500
3rd year	348	470	1000	940	260	730	800	260	1000
4th year	522	700	1500	1410	390	1095	1200	390	1500
5th year	696	940	2000	1880	520	1460	1600	520	2000

Under irrigated conditions, the fertilizers can be applied in 3-4 equal split doses.

1000:500:2000gm NPK/ palm/year is to be recommended for **hybrid coconut** in coastal Tamil Nadu.

Fertigation may be done at monthly intervals with 75% of the recommended dose of the above fertilizers.

Recommended Micro Nutrient Mixture – Tamil Nadu

Quantity recommended: 1 kg/palm/year (Ferrous, Manganese, Zinc, Boron, Copper)

Method of application: Soil application (Basal) apply along with farm yard manure.

[↑Top of page](#)

Manuring in Karnataka

Organic Matter: FYM/ Compost: Before Planting – 12.5 t/ha. Every Year – 50 kg/palm

Fertilizers Requirement for Pre-monsoon Condition

Time after planting	Blanket Recommendation (g/palm/year)						Straight Fertilizer (g/palm/year)					
	Pre monsoon			Post Monsoon			Pre monsoon			Post Monsoon		
	N	P	K	N	P	K	Urea	SSP	MOP	Urea	SSP	MOP
1st year after planting	Planting in May – June			50	40	135	-	-	-	110	250	225
2nd year after planting	50	40	135	110	80	270	110	250	225	240	500	450
3rd year after planting	110	80	270	220	160	540	240	500	450	480	1000	900
4th year after planting	170	120	400	330	200	800	370	750	670	720	1250	1335

Time after planting	DAP & Others (g/palm)					
	Pre monsoon			Post monsoon		
	DAP	Urea	MOP	DAP	Urea	MOP
1st year after planting	-	-	-	87	75	225
2nd year after planting	87	75	225	175	170	450
3rd year after planting	175	170	450	348	340	900
4th year after planting	260	267	670	435	545	1335

Time after planting	17:17:17: Complex (g/palm)					
	Pre monsoon			Post monsoon		
	17:17:17 Complex	Urea	MOP	17:17:17 Complex	Urea	MOP
1st year after planting	-	-	-	235	22	158
2nd year after planting	235	22	158	470	65	315
3rd year after planting	470	65	315	941	130	630
4th year after planting	705	110	465	1176	282	664

Time after planting	20:20:0 complex (g/palm)					
	Pre monsoon			Post monsoon		
	20:20:0 complex	Urea	MOP	20:20:0 complex	Urea	MOP
1st year after planting	-	-	-	200	22	225
2nd year after planting	200	22	225	400	65	450
3rd year after planting	400	65	450	800	130	900
4th year after planting	600	110	670	1000	282	1335

Apply MgSO₄ for bearing trees @ 0.5 kg/palm in coastal region (Note: Lime incorporation – 15 days prior to application of fertilizer during September – October.

[↑Top of page](#)

Manuring in Kerala

Manuring young palms – KAU: For the first two years after planting under rainfed conditions, apply fertilizers in two split doses at the rates shown in the table given below

Fertilizer Requirement of Young Palms in Relation to that of Adult Palms– KERALA (Under Rainfed Condition)

Time after planting	Blanket Recommendation (g/palm)			Straight Fertilizer (g/palm)		
	N	P	K	Urea	Super Phosphate	Muriate of potash
3 months	50	32	120	110	200	200
1st year	167	107	400	360	670	660
2nd year	334	214	800	720	1340	1330

Time after planting	DAP & Others (g/palm)			17:17:17: Complex (g/palm)			20:20:0 complex (g/palm)		
	DAP	Urea	MOP	17:17:17 Complex	Urea	MOP	20:20:20 complex	Urea	MOP
3 months	70	81	200	188	40	146	160	40	200
1st year	232	271	660	630	130	486	535	130	660
2nd year	464	545	1330	1258	260	973	1070	260	1330

Note: Under irrigated conditions, the fertilizers can be applied in 3-4 equal split doses. In the case of low lying areas, apply fertilizer after water table recedes in one single dose or in two split doses as conditions permit. In all types of soils that are low in organic matter content (except reclaimed clayey soils and alluvial soils), apply organic matter @ of 15-25 kg/palm/year during June-July from the second year of planting.

[↑Top of page](#)

Fertilizer Requirement of Adult Palms– KERALA (Under Rainfed Condition)

	General Recommendation (g/palm)	Straight Fertilizer (g/palm)

	N	P	K	Urea	Amm. Sulphate	Super Phosphate	Rock Phosphate	Muriate of Potash
(a) Average management	340	170	680	750	1650	1060	570	1130
(b) Good management	500	320	1200	1080	2000	2000	1070	1990
For reclaimed clayey soils (as in Kuttanad)	250	350	900	540	1190	2190	1170	1495
Red loam soils (southern Kerala)	680	230	900	1480	3240	1440	770	1495
Hybrids & High Yielding Palms								
(a) For irrigated areas	1000	500	2000	2170	4880	3130	1670	3320
(b) For rain fed conditions	500	320	1200	1090	2380	2000	1070	1990

Time After Planting	DAP & Others (g/palm)			17:17:17: Complex (g/palm)			20:20:0 Complex (g/palm)		
	DAP	Urea	MOP	17:17:17 Complex	Urea	MOP	20:20:0 Complex	Urea	MOP
(a) Average management	370	595	1130	1000	370	847	850	370	1130
(b) Good management	695	814	1990	1882	390	1460	1600	390	1990
For reclaimed clayey soils (as in Kuttanad)	760	297	1495	1470	625 (SSP)	1080	1250	625 (SSP)	1495
Red loam soils (southern Kerala)	500	1280	1495	1353	977	1112	1150	977	1495
Hybrids & High Yielding Palms									

(a) For irrigated areas	1085	1750	3320	2940	1085	2490	2500	1085	3320
(b) For rain fed conditions	695	814	1990	1882	390	1460	1600	390	1990

1. Under irrigated conditions, fertilizers can be applied in 3-4 equal split doses.
2. In the case of low-lying areas, apply fertilizers in one single dose after water table recedes or in two split doses as conditions permit.
3. The application of organic materials such as forest leaves, cattle manure, coir dust or coconut shredding at 10 kg per pit in the first three years and 15-25 kg thereafter will be useful to obtain better establishment of coconut palms in sandy soils and in coastal situations.
4. In situations where the available P₂O₅ status of the soil is more than 10 ppm, application of phosphatic fertilizers can be skipped for a few years until the status of P₂O₅ reaches 10 ppm.
5. For sandy and sandy loams of Onattukara and similar situations and also for hybrid palms grown in root (wilt) affected areas, apply 500 g N + 300 g P₂O₅ + 1000 g K₂O along with 500 g MgSO₄ / palm / year.
6. Application of MgSO₄ to coconut palms earlier confined to root wilt affected areas is recommended for the whole state (ad hoc recommendation).
7. The N:P₂O₅:K₂O recommendation given for high yielding palms is, in general, sufficient for palms yielding up to 100 nuts per year. For palms yielding more than 100 nuts per year, an additional dose of 10 g N, 5 g P₂O₅ and 15 g K₂O may be supplied for every nut exceeding 100 nuts (ad hoc recommendation).

In laterite soils, 50% of the K₂O requirement of coconut can be substituted by Na₂O supplied in the form of sodium chloride.

[↑Top of page](#)

Time, Frequency and Method of Fertilizer Application: The proper time for the application of fertilizers is when sufficient moisture is available in the soil. Under **rainfed** conditions, apply fertilizers in two split doses, 1/3 at the time of early southwest monsoon showers in April-June and 2/3 in September-October. Under **irrigated** conditions, apply fertilizers in three or four equal doses in April-May, August-September, December and February-March.

Apply lime or dolomite during April-May, magnesium sulphate during August-September and organic matter during June-July. For an adult palm 1 kg dolomite or 1 kg lime + 0.5 kg MgSO₄ is required per annum.

Apply fertilizers and manures in circular basins at a radius of 2.0 m from the base of the palm and 10 cm deep, opened after the onset of southwest monsoon. Split doses can be applied with irrigation water in summer months.

Application of MgSO₄ at the rate of 500g/palm/year for adult coconut plantations is recommended.

Boron deficiency causes characteristic malformation of leaves like hook leaves, nut cracking, drying of the female flowers etc. Soil application of Borax @ 50 g/tree twice at monthly intervals after appearance of the first symptom corrects the deficiency. In the root (wilt) disease affected area, it has been recommended to apply Borax @ 300 g/seedling and Borax @ 500 g/ adult tree. Application of magnesium @ 500 g MgO per palm is advantageous for the management of root (wilt) diseased palms to restore palm vigour and sustain the productivity.

[↑Top of page](#)

Biofertilizers: The root regions of coconut palm are inhabited by a number of free living and associative symbiotic nitrogen-fixing bacteria having nitrogenase activity. The association of N₂-fixing *Azospirillum amazonense* with the roots of coconut palm has recently been confirmed. The other associative nitrogen fixers include *Azospirillum lipoferum*, *Azospirillum brasilense*, *Herbaspirillum frisingense*, *Bacillus* spp., *Burkholderia* spp., *Azoarcus* spp., *Arthrobacter* spp. and many more which are yet to be identified. Some of these are capable of fixing nitrogen even in the presence of nitrogen fertilizer.

Microbial combinations may have very great potential for plant growth enhancement. While raising coconut seedlings in coir dust-soil mixture, *Beijerinckia indica* and associative diazotrophs such as *Azospirillum*, *Arthrobacter*, *Azoarcus*, *Herbaspirillum*, *Bacillus*, *Burkholderia* and *Pseudomonas* are promising microbiological inoculants which enhances the growth and performance of seedlings.

[↑Top of page](#)

Vermicomposting: The local species of *Eudrilus* identified from coconut garden which is superior to other species in composting of coconut plantation wastes. This earthworm can be multiplied fast in a 1:1 mixture of cowdung and decayed organic wastes. Keep 10 kg of this mixture in bucket and release 50 to 100 worms. Mulch the mixture with grass and cover with a net. Maintain the moisture. In 1 to 2 months 150 to 300 g earthworm will be produced.

[↑Top of page](#)

Coirpith Composting: Coir pith contains very less nitrogen and has large amounts of lignin and phytotoxic polyphenols and has to be composted before using as manure. Exposure to rains and sunlight for many years results in loss of problematic chemicals and the use of weathered coir pith may be advantages. Fresh coir pith has a wide C:N ratio (about 100:1) and for initiating microbial action, nitrogenous organic or inorganic materials are to be added. Additionally, fortification with rock phosphate at the rate of 10kg per ton of coir pith can also favor microbial action. The well known technology for composting coir pith using *Pleurotus sajor caju* may be utilized for large scale composting.

For composting one ton of coir pith, 5kg of urea and 5 bottles of *Pleurotus* spawn is required. 100 kg of coir pith is spread on a level land in shaded place and 1 bottle of spawn is sprinkled over it. The spawn layer is covered with 100 kg coir pith and 1 kg urea is sprinkled over it. This process is repeated 5 times to get a heap, which is protected from direct sunlight and rain. Proper moisture to be maintains in the heap and is allowed to undergo degradation for 1 month. This compost can be used as manure in coconut plantations.

[↑Top of page](#)

Recycling of Palm Waste: Recycling of palm waste is very much beneficial especially for maintaining the availability status of micronutrients and trace elements. Palm wastes like coconut leaves, crown waste, dried spathes, husk etc. may be deposited in a small trench of convenient length, 0.5 m to 0.75 m wide and 0.3 to 0.5 m deep at a distance of 2-2.5 m away from the base of the trunk. Fill up this trench with the palm wastes along one side of the palm (say north) in one year, opposite side (south) in the next year, east in the third year and so on. This practice of organic recycling of waste has been found to improve the growth and productivity of the palms.

[↑Top of page](#)

TNAU Coconut Tonic: For nut bearing coconut, root feed TNAU coconut tonic @ 200ml/palm once in six months.

[↑Top of page](#)

Bio-Fertilizer Recommendation

1. 50 g of *Azospirillum*
2. 50 g of *Phosphobacteria* (or) 100 g *Azophos*
3. 50 g of VAM

Mix all the contents in sufficient quantity of compost or FYM and apply near feeding roots once in 6 months / palm starting from planting. Don't mix with chemical fertilizers and pesticides

[↑Top of page](#)

Organic Recycling: Any one of the green manure crops like sunnhemp, wild indigo, Calapagonium or Daincha may be sown and ploughed *in situ* at the time of flowering as a substitute of compost to be applied. Sow sunnhemp @ 50 g/palm in the basin and incorporate before flowering. Coir pith compost/vermicompost made from coir pith/ coconut leaves/ other wastes from coconut grove can be applied.

[↑Top of page](#)

Physiological Disorder

Deficiencies for nutrients such as nitrogen, phosphorus, Potassium and micronutrients are very common in coconuts cultivation. Symptoms are often observed in coconut gardens where the trees are not properly fertilized especially with organic manures.

(i) Major Deficiency

Nitrogen (N) Deficiency

Symptoms: N deficiency is typically caused by insufficient N in the soil. Nitrogen deficiency begins as a uniform light green discoloration / yellowing (uniform chlorosis) of the oldest leaves. *Yellowing starts from tip to base of the lower leaves and will proceed up.* As the deficiency progresses, younger leaves will also become discoloured. *Older leaves are golden yellow colour.* Growth virtually stops when N deficiency is severe and become shedding of leaves.

Diagnostic Techniques: Nitrogen deficiency is easily diagnosed by symptoms alone, although leaf nutrient analysis can also be helpful. Nitrogen deficiency can be confused with Fe or S deficiency, although the chlorosis in those deficiencies is typically most severe on the youngest leaves. The reverse is true for N deficiency.

Management: Foliar application of 2% urea thrice at fortnightly interval or soil application of 1-2 kg urea / tree or Root feeding of 1% urea 200 ml twice a year.

[↑Top of page](#)

Phosphorus Deficiency

Symptoms of damage: Deficiency occurs in acid and alkaline soils.

- Purple coloration in leaves (In severe cases may leaves turn yellow before drying prematurely)
- Sluggish growth
- Leaves stay upright

- Premature leaf shedding

The growth, leaf size and leaf number reduced. The root growth is restricted if phosphorus deficiency is recorded. There are no clear visual symptoms of phosphorus deficiency other than stunting and decrease in yield. This deficiency common in wide range of soil

Management: Foliar spray of DAP 2% twice at fortnightly interval or soil application of FYM 5kg/tree. Root feeding of 1% DAP 2 ml twice a year.

[↑Top of page](#)

Boron (B) Deficiency

Symptoms: Boron deficiency is caused by insufficient B in the soil. It may be caused by soil drying and high soil pH, while temporary B deficiency is caused by heavy leaching.

- Symptoms always occur on newly emerging leaves, and remain visible on these leaves as they mature and are replaced by younger leaves.
- One of the earliest symptoms of B deficiency on coconut palm is leaf wrinkling and manifested as sharply bent leaflet tips, commonly called “hook leaf”. These sharp leaflet hooks are quite rigid and cannot be straightened out without tearing the leaflets. Leaves have a serrated zigzag appearance.
- One of the most common symptoms of B deficiency is the failure of newly emerging spear leaves to open normally. In a chronic stage, multiple unopened spear leaves may be visible at the apex of the canopy.
- Boron deficiency also occurs in inflorescence and nuts. The inflorescence and nuts are become necrotic.

Diagnostic Techniques: Boron deficiency symptoms are quite distinctive and are usually sufficient for diagnosis by themselves. Manganese deficiency produces symptoms similar to those of B deficiency, but no other common deficiency produces symptoms that could be confused with those of B deficiency. Because B deficiency is so transient in nature, the element is immobile within the palm (cannot move from one leaf to another), and deficiencies affect only leaf primordial developing within the bud area, leaf analysis is not particularly useful.

Application of Boron to Coconut			
Stage	Source (choose one)	Rate	Time and Method
Nursery	Borax/Sodium Tetraborate	0.2% (2 g/L H ₂ O), (75-100 ml/seedling)	1-2 times by spraying or drenching

1 year old	Borax/Sodium Tetraborate/Octaborate	5-10 g/plant	Once each year to soil
2-3 years old	Borax/Sodium Tetraborate/Octaborate	15-20 g/plant	Once each year to soil
4 years old and more	Borax/Sodium Tetraborate/Octaborate	30-50 g/plant	Once every 2 years by soil

Popularity: 4%

[↑Top of page](#)

Manganese (Mn) Deficiency: Manganese deficiency is very common on alkaline soils, but can occur in containers if drainage is poor or soils temperatures are cool.

Symptoms: The newest leaves of Mn deficient palms emerge chlorotic with longitudinal necrotic streaks. As the deficiency progresses, newly emerging leaflets appear necrotic and withered on all but basal portions of the leaflets. This withering results in a curling of the leaflets about the rachis giving the leaf a frizzled appearance ('frizzle top'). On new leaves of Mn-deficient palm, necrotic leaflet tips fall off and the leaf has a signed appearance. In severely Mn- deficient palms, growth stops and newly emerging leaves consist solely of necrotic petiole stubs.

Diagnostic Techniques: Visual symptoms may be sufficient to diagnose this disorder, but leaf nutrient analysis is also suggested, since symptoms of boron (B) deficiency can be similar. Late stage potassium (K) deficiency symptoms are virtually indistinguishable from those of Mn deficiency at a distance and close examination is required to look for characteristic longitudinal streaking and basal symptom distribution of Mn deficiency.

Management: Soil application of MnSO₄ @ 25kg/ha

[↑Top of page](#)

Magnesium (Mg) Deficiency: Magnesium deficiency is caused by insufficient Mg in the soil. Magnesium is readily leached from sandy soils and other soils having little cation exchange capacity. High levels of nitrogen (N), Potassium (K), or calcium (Ca) in the soil can also induce or exacerbate Mg deficiencies.

Symptoms: Magnesium deficiency appears on the oldest leaves of palms as broad chlorotic (yellow) bands along the margins with the central portion of the leaves remaining distinctly green. In severe cases leaflet tips may become necrotic. Older leaves become bronzed and dry appearance. Leaflets show necrosis and turn to reddish brown with translucent spots

yellowing starts at the tip and spreads to the base.

Diagnostic Techniques: Visual symptoms alone are usually sufficient to diagnose Mg deficiency. Magnesium deficiency symptoms differ from those of K deficiency in that symptom severity of discoloration K- deficient leaves is usually orange to bronze, shading gradually to green at the base of the leaf, whereas Mg- deficient leaves have distinctly green leaf centers and bright lemon yellow to orange margins.

Management: Soil application of MgSO_4 1-2 kg/tree/year. Root feeding of 200 ml of 0.2% MgSO_4 twice a year.

[↑Top of page](#)

(ii) Minor Deficiency

Sulphur Deficiency

Symptoms: Typical symptoms are yellowish-green or yellowish-orange leaflets. Older leaves are remaining green. Leaves droop as the stem becomes weak. In older palms, leaf number and size are reduced. Sometimes an apron of dead fronds develops around the stem due to weakness of the rachis. Nuts may fall prematurely. Copra is rubbery and of poor market quality.

Management: Soil application of gypsum 2 - 5 kg/tree/year. Root feeding of 0.2% gypsum.

[↑Top of page](#)

Iron Deficiency

Symptoms: Iron deficiency usually appears on palms growing in poorly aerated soils or those that have been planted too deeply. Water logged soils and deep planting effectively suffocate the roots and reduce their effectiveness in taking up nutrients such as Fe. The main symptom of iron deficiency is chlorosis or yellowing between the veins of new leaves (Uniform chlorotic new leaves as the deficiency progresses, the tips become necrotic and leaf size reduced).

Management: Application of FeSO_4 0.25 to 0.5 kg/tree/year

[↑Top of page](#)

Zinc Deficiency

Symptoms: Zinc deficiency is characterized by formation of small leaves wherein the leaf size is reduced to 50%. Leaflets become chlorotic, narrow and reduced in length. In acute deficiency, flowering is delayed. Zinc deficiency will also lead to button shedding. It occurs mostly in saline soils.

Management: Soil application of ZnSO₄ @ 25kg/ha

[↑Top of page](#)

Calcium Deficiency

Symptoms: Young leaves exhibit narrow white bands at margins. Interveinal chlorosis. Rusty appearance in leaf margin. Rolling up of leaves. Occurs only in acid soil

Management: Soil application of lime based on lime requirement and root feeding of 1% calcium nitrate.

[↑Top of page](#)

Copper Deficiency

Symptoms: Coppery bluish leaf. Rolling of terminal leaves due to loss of turgor. Leaves appear to be bleached grey. Fail to produce flowers

Management: Soil application of CuSO₄ @ 25 kg per ha.

[↑Top of page](#)

Protection technologies of Coconut: Management practices include growing green manure (sun hemp) as an intercrop and ploughing in situ. Application of 650g urea, 1kg super phosphate and 1 kg potash per tree along with 25 kg farmyard manure (FYM) once in six months in June- July and December- January. Application of micronutrients viz., magnesium sulphate 150g, zinc sulphate 75g, borax 50g along with 10kg of well decomposed FYM per tree prior to rainy season. These measures help overcome nutritional disorders and restore the vigour and productivity of the trees.